

Landslide Segmentation

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1 Introduction

Landslides are a frequent and destructive natural hazard, particularly in mountainous terrain. Large-scale events can damage critical infrastructure and result in significant loss of life. In the immediate aftermath of an event, timely and consistent mapping of the affected areas is a vital component of risk assessment and emergency response.

Historically, this mapping has been conducted through time-consuming manual delineation from aerial or satellite imagery, a process that is both subjective and difficult to scale during a crisis. In recent years, data-driven methods using deep learning have emerged as a powerful alternative. Semantic segmentation models, such as U-Net and Transformer-based architectures, can learn to identify complex landslide patterns from high-resolution imagery, offering a path to automated, large-scale mapping.

However, a critical challenge remains: **generalization**. A model trained on the geological and sensor characteristics of one region often fails dramatically when applied to another. This *domain shift*—caused by differences in terrain, vegetation, sensor types, and image resolution—is the primary barrier to creating a truly operational tool.

This study directly confronts this challenge. Our contribution is a rigorous, empirical evaluation of modern segmentation models under a strict **Leave-One-Out (LOO)** protocol. By holding out an entire geographic region from the training process, we simulate a real-world deployment scenario. We specifically investigate two key factors:

1. **Architecture:** We compare the performance of the classic U-Net against the modern Transformer-based Segformer.
2. **Class Imbalance:** Given that landslides comprise only approximately 2% of the pixels, we systematically evaluate advanced loss functions (e.g., BCE+Dice, Focal) designed to handle this extreme imbalance.

This document reports our findings and identifies the most promising model–loss combination for building a robust, generalizable landslide segmentation tool.

2 Business Understanding

The primary objective of this project is to assess the feasibility of creating a **single, robust model for rapid landslide mapping** that can be reliably deployed in a *new* geographical region *after* a disaster event.

For such a tool to have real-world value for organizations involved in disaster response (e.g., civil protection agencies, geological surveys), it cannot require site-specific re-training, for which there would be neither sufficient time nor data. The model must function *off-the-shelf* in an area it has never seen before.

The **core business challenge**, therefore, is **domain shift**. The heterogeneity across the 22 available regions—in terms of resolution, sensor characteristics, and geomorphology—is not

a limitation but the central feature of the problem. A model that achieves a high score on a simple test split but fails when applied to a new region represents a business failure.

To address this challenge, our entire methodology is built around the **Leave-One-Out (LOO)** evaluation protocol. This protocol directly simulates the real-world deployment scenario: for each of the 22 regions, we train a model on the *other 21* and test it on the *one held-out region*.

Success is therefore measured not by the highest achievable F1 score in a single area, but by the **highest average F1 score** and the **lowest performance variance** across all 22 LOO runs. These metrics indicate whether a model can generalize beyond its training data and serve as a truly useful, deployable asset.

Our investigation into different architectures (U-Net vs. Segformer) and loss functions is guided by this ultimate goal: identifying the most generalizable solution for large-scale, post-disaster landslide mapping.

3 Data Understanding

The dataset covers 22 geographic regions. Each region includes a high-resolution image and a corresponding landslide mask. Key properties:

- Class imbalance: approximately 98% of pixels are non-landslide.
- Landslide objects: small, thin, and irregular, challenging for pixel-wise learning.
- Heterogeneity: regions differ in sensor characteristics and resolution per pixel.



(a) Example Image Patch



(b) Corresponding Mask

Figure 1: Example of a 512×512 patch and its binary landslide mask, illustrating the sparse and irregular nature of the positive class.

4 Data Preparation

To enable mini-batch training, each large image was split into 512×512 patches. This is a common strategy but may discard global context that could help classification near large-scale geomorphic patterns. All image patches were normalized and standardized prior to model input: pixel intensities were scaled to $[0,1]$ and standardized using per-channel mean and standard deviation computed over the training set. No test data were used during training or model selection.

5 Modeling

We adopted a Leave-One-Region-Out protocol to evaluate generalization under domain shift. For each run, one region served as the test set; the remaining regions formed the training/validation sets. No self-training was performed, and test regions were never used for weight updates.

We evaluated U-Net and Segformer (b0 and b1) architectures. Given the imbalance and object sparsity, we focused on losses commonly used for segmentation:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (1)$$

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N} \sum_{i=1}^N [w y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad w = \frac{N_{\text{neg}}}{N_{\text{pos}}} \approx 49, \quad (2)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2|Y \cap \hat{Y}| + \epsilon}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}, \quad (3)$$

$$\mathcal{L}_{\text{Focal}} = -\alpha(1 - p_t)^\gamma \log(p_t), \quad p_t = \begin{cases} \hat{y} & y = 1 \\ 1 - \hat{y} & y = 0 \end{cases}, \quad \alpha = 0.75, \gamma = 2.0, \quad (4)$$

$$\mathcal{L}_{\text{Lovász}} = \frac{1}{N} \sum_{i=1}^N \overline{\Delta J}(m_i), \quad m_i = 1 - y_i f_i. \quad (5)$$

Dice and Lovász losses were used in combination with BCE.

Metrics. We report pixel and patch-level F1 scores, AUC-ROC, and validation loss (Binary Cross-Entropy) for comparison across settings.

6 Evaluation

6.1 Loss Function Comparison (U-Net baseline)

Table 1 summarizes average performance across regions for different losses with U-Net. BCE+Dice and Focal losses yield higher F1 scores than plain BCE, while Weighted BCE improves validation loss.

Table 1: Loss comparison (mean $\pm$ std).

Criterion	Valid Loss	F1 (pixel)	F1 (patch)	AUC-ROC
BCE	0.1329 ± 0.0161	0.2563 ± 0.0597	0.4917 ± 0.0706	0.6605 ± 0.0625
WBCE	0.1170 ± 0.0092	0.2673 ± 0.0632	0.5139 ± 0.0644	0.7014 ± 0.0418
BCE+Dice	0.1216 ± 0.0088	0.3206 ± 0.0666	0.5803 ± 0.0649	0.6984 ± 0.0456
BCE+Lovász	0.1224 ± 0.0079	0.2563 ± 0.0695	0.3831 ± 0.1171	0.7299 ± 0.0349
Focal	0.1401 ± 0.0089	0.3204 ± 0.0593	0.5835 ± 0.0722	0.7272 ± 0.0339

Based on these results, subsequent experiments prioritize BCE+Dice and Focal losses. Empirically, Focal showed (i) smoother validation loss trends toward minima, (ii) more stable gradient norms during training, and (iii) comparatively high AUC-ROC. Figures 2, 3, 4, and 5 visualize these trends for the U-Net baseline under each loss.

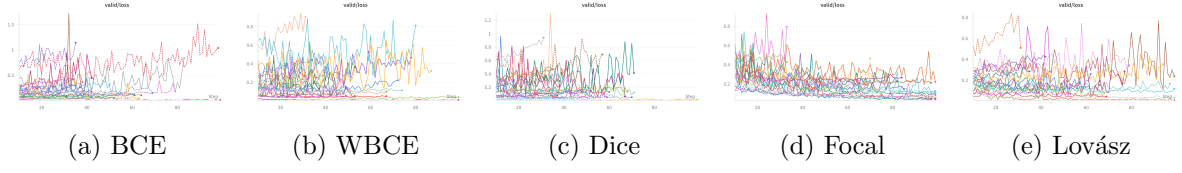


Figure 2: Validation loss across epochs for U-Net under different losses.

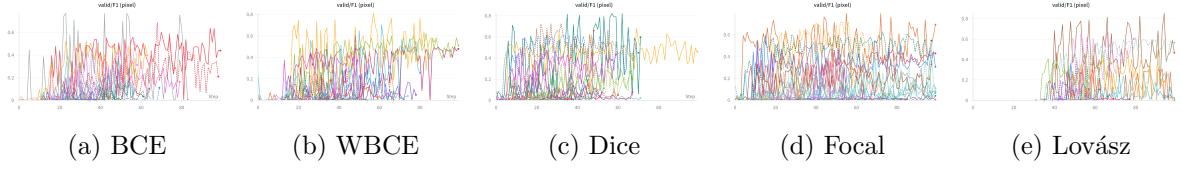


Figure 3: Validation F1 across epochs for U-Net under different losses.

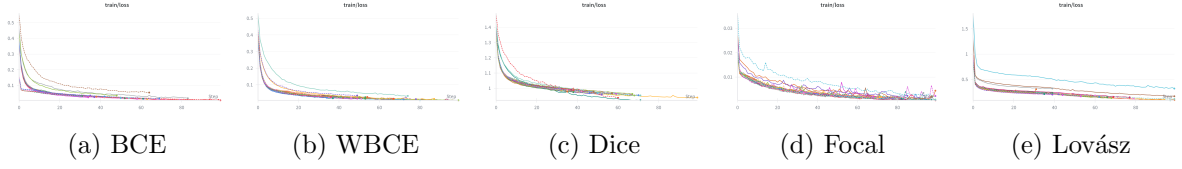


Figure 4: Training loss across epochs for U-Net under different losses.



Figure 5: Training gradient norm (proxy) across epochs for U-Net under different losses.

These behaviors are consistent with the theoretical properties of the Focal loss. By down-weighting well-classified examples and emphasizing harder or misclassified pixels, the loss reduces the overwhelming influence of dominant negative samples in highly imbalanced data. This focus helps stabilize gradient updates, encouraging the network to learn more discriminative features for rare landslide pixels. As a result, convergence tends to be smoother and the model achieves stronger recall and overall F1 performance without significantly degrading precision.

6.2 Best Model per Test Region

Table 2 lists the best model per region for BCE+Dice and Focal losses based on pixel-wise F1. In many regions, Segformer-b0 emerges as the top performer; U-Net remains competitive in a subset of regions.

Table 2: Best model per test region (by pixel-wise F1). Bold indicates the higher F1 between the two losses for that region.

Dataset	BCE+Dice	Focal
A19_5cm	L4S/unet	L4S/unet
Avigliano2_5_cm	nvidia/segformer-b0	nvidia/segformer-b0
Avigliano_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Bella3_5cm	L4S/unet	L4S/unet
Bella_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Corleto.Perticara_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Ferrandina2_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Genzano.Lucania_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Grassano_5cm	L4S/unet	nvidia/segformer-b0
Ischia_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Miglionico_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Montescaglioso_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Pietragalla_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Pomarico_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Provincia_Taranto_Ginosa_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Provincia_Taranto_Montemesola_5cm_1	nvidia/segformer-b0	nvidia/segformer-b0
Provincia_Taranto_Montemesola_5cm_2	nvidia/segformer-b0	nvidia/segformer-b0
Provincia_Taranto_Montemesola_5cm_3	L4S/unet	nvidia/segformer-b0
Stigliano2_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Stigliano3_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Stigliano4_5cm	nvidia/segformer-b0	nvidia/segformer-b0
Stigliano_5cm	nvidia/segformer-b0	nvidia/segformer-b0

6.3 Segformer-b0 vs. Segformer-b1 (Focal loss)

Table 3 reports average metrics. Segformer-b1 shows slightly lower validation loss, whereas Segformer-b0 achieves higher F1 scores on average.

Table 3: Overall performance with Focal loss.

Model	Avg. Valid Loss	Avg. F1 (pixel)	Avg. F1 (patch)	Avg. AUC-ROC
Segformer-b0	0.0957	0.5045	0.6886	0.8532
Segformer-b1	0.0927	0.4620	0.6664	0.8533

Per-region pixel-wise F1 comparison for Segformer-b0 and b1 is shown in Table 4. Positive Δ indicates $b1 > b0$.

Table 4: Per-region F1 (pixel) comparison: Segformer-b0 vs. b1.

Dataset	Segformer-b0	Segformer-b1	$\Delta F1$ (b1-b0)
A19_5cm	0.3656	0.3051	-0.0605
Avigliano2_5cm	0.1490	0.1742	+0.0252
Avigliano_5cm	0.8315	0.7772	-0.0543
Bella3_5cm	0.5599	0.5982	+0.0383
Bella_5cm	0.2561	0.2422	-0.0139
Corleto_Perticara_5cm	0.6294	0.4604	-0.1691
Ferrandina2_5cm	0.6227	0.6144	-0.0083
Genzano_Lucania_5cm	0.6067	0.4998	-0.1069
Grassano_5cm	0.5223	0.3729	-0.1494
Ischia_5cm	0.4124	0.2239	-0.1885
Miglionico_5cm	0.6518	0.7184	+0.0666
Montescaglioso_5cm	0.6778	0.6537	-0.0241
Pietragalla_5cm	0.8480	0.8279	-0.0200
Pomarico_5cm	0.2406	0.1474	-0.0932
Provincia_Taranto_Ginosa_5cm	0.2338	0.2085	-0.0253
Provincia_Taranto_Montemesola_5cm_1	0.5991	0.5655	-0.0336
Provincia_Taranto_Montemesola_5cm_2	0.5975	0.5726	-0.0249
Provincia_Taranto_Montemesola_5cm_3	0.6876	0.6243	-0.0633
Stigliano2_5cm	0.4434	0.4045	-0.0390
Stigliano3_5cm	0.1723	0.1781	+0.0058
Stigliano4_5cm	0.4584	0.4678	+0.0094
Stigliano_5cm	0.5326	0.5279	-0.0047

7 Conclusion

Under the chosen protocol, Segformer-b0 presents a favorable trade-off between accuracy and complexity. Segformer-b1 (13.7M parameters) attains slightly lower validation loss but does not translate this into higher F1 on average compared to Segformer-b0 (3.7M parameters).

A likely explanation for this is that the larger Segformer-b1 model is more prone to overfitting on the 21 training regions, harming its ability to generalize to the held-out test region. The lighter Segformer-b0 is effectively regularized by its smaller capacity, forcing it to learn more robust and generalizable features. This result strongly suggests that for this domain generalization task, larger model capacity may be detrimental.

Furthermore, while Segformer-b0 was the top performer on average, Table 2 shows that the U-Net baseline remained the best-performing model in 4 of the 22 regions under the Focal loss. This indicates that the strong convolutional inductive bias of U-Net is still highly effective for certain specific geomorphological and sensor domains.

These observations are specific to the dataset, preprocessing, and training settings used here.

Model	Parameters
L4S/unet (U-Net)	17,267,393
nvidia/segformer-b0	3,714,401
nvidia/segformer-b1	13,677,505

This study evaluated the generalization performance of segmentation models for landslide mapping using a strict Leave-One-Region-Out protocol. We found that addressing the severe class imbalance is critical; **BCE+Dice** and **Focal** losses significantly outperformed standard BCE, with Focal loss providing stable training and high F1 and AUC-ROC scores. When comparing architectures, the Transformer-based **Segformer-b0** demonstrated the best overall generalization, achieving a higher average pixel-wise F1 score (0.5045) than the larger Segformer-b1 (0.4620) and the U-Net baseline. Notably, Segformer-b0 is also the most lightweight model tested (3.7M parameters), making it a compelling choice for a robust, efficient, and deployable tool for post-disaster rapid mapping.

8 Limitations and Future Work

The results presented reflect the specific dataset, 512×512 patching strategy, and hyperparameters used. We did not explore domain adaptation or multi-scale context aggregation beyond the tested architectures.

Based on these findings, future work should focus on:

- Data regularization and more advanced augmentation strategies to further combat overfitting.
- Exploring if finer-grained annotation could improve supervision on narrow landslide features.
- Incorporating additional training data from geographically distinct regions to further improve generalization.
- Investigating patch-wise objectives